

CAMBRIDGE INTERNATIONAL AS LEVEL BIOLOGY (9700)

TOPIC 7 – TRANSPORT IN PLANTS

Vascular Histology, Cohesion–Tension Transpiration, Xeromorphic Adaptations & Phloem Mass Flow

CANDIDATE

Hamna (Grade A* Scholar)

SPECIFICATION

Cambridge 9700 AS Level

TOTAL MARKS / QUESTIONS

220 Marks · 50 Questions

EXAMINATION STRUCTURE & 80/20 QUESTION TARIFF

- Section A – Extended High-Tariff Questions: 20 × 6 Marks (120 Marks total) [Multi-part synoptic evaluation]
- Section B – Data Interpretation & Structured Explanations: 20 × 4 Marks (80 Marks total)
- Section C – Core Definitions & Quantitative Calculations: 10 × 2 Marks (20 Marks total)
- Section D – 10 High-Frequency Core Repeated Questions for Exam Mastery & Knowledge Consolidation
- Division: 80% Authentic Cambridge 9700 Past Paper references + 20% Original A* Extension Problems

SYLLABUS SUBTOPICS COVERED IN THIS PACK

7.1 Structure of transport tissues: plan diagrams of transverse sections (TS) of stems, roots, and leaves of herbaceous dicotyledonous plants; precise spatial distribution of xylem and phloem in stem (peripheral ring, internal xylem), root (central stele, star/cross-shaped xylem), and leaf midrib (upper/adaxial xylem, lower/abaxial phloem); ultrastructure and cellular adaptations of xylem vessel elements (lignified secondary walls, hollow lumen from autolysis of protoplast, absent/perforated end walls, bordered pits for lateral bypass); ultrastructure and interdependence of phloem sieve tube elements (reduced peripheral cytoplasm, absence of nucleus, vacuole, ribosomes, perforated sieve plates with callose-lined pores) and companion cells (prominent nucleus, abundant mitochondria, dense ribosomes, extensive plasmodesmata, transfer cell wall ingrowths).

7.2 Transport mechanisms: uptake and transport of dissolved mineral ions and water across root cortex; apoplast pathway (porous cellulose cell walls and intercellular spaces) vs symplast pathway (living cytoplasm interconnected via plasmodesmata); endodermal checkpoint and the role of the suberised, hydrophobic Casparian strip in forcing apoplastic water into symplast for selective membrane filtration; transpiration as the loss of water vapour from aerial parts of the plant via evaporation from wet mesophyll cell walls into intercellular spaces followed by diffusion through stomata down water potential gradient; cohesion–tension theory (transpiration pull/tension, hydrogen bonding cohesion of unbroken water column, adhesion to hydrophilic cellulose and lignin); experimental investigation of water uptake using calibrated potometers; xerophytic leaf adaptations (sunken stomata, epidermal trichomes, rolled leaves in *Ammophila arenaria*, thick waxy cuticle); translocation of sucrose and amino acids from sources to sinks via phloem; chemiosmotic phloem loading in companion cells via H^+ -ATPase proton pumps and secondary active H^+ /sucrose cotransporters; mass flow hypothesis driven down hydrostatic pressure gradients (ΔP) generated by osmotic water movement at source and sink; classical aphid stylet and girdling evidence.

EXAMINER'S DIRECTIVE FOR CANDIDATE HAMNA

Write legibly in the dotted spaces provided. Pay strict attention to Cambridge command words (Explain, Describe, Deduce, Calculate).

Where biological diagrams or graphs are provided, extract exact qualitative and quantitative coordinates to support your answers.

1 9700/22/M/J/23/Q3 – Anatomical Distribution of Vascular Tissues in Dicot Stem and Root

[6 Marks] Syllabus 7.1

The spatial arrangement of primary vascular tissues differs characteristically between the stems and roots of herbaceous dicotyledonous plants. Fig. 7.1 shows plan diagrams of a transverse section (TS) through a dicot stem and a TS through a dicot root.

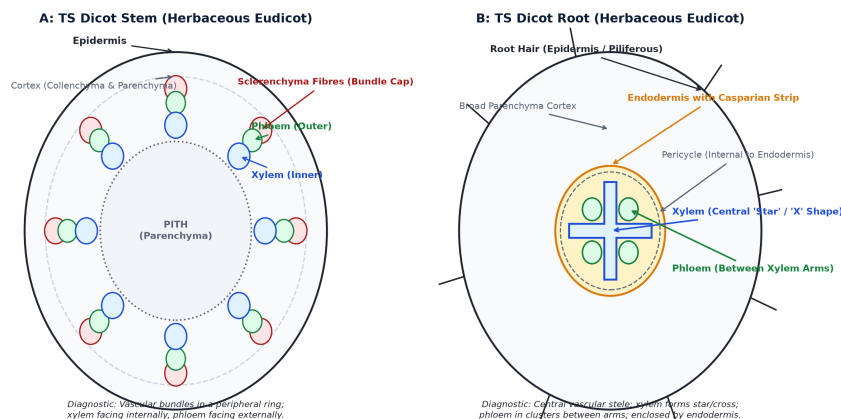


Fig. 7.1: Low-power tissue plan diagrams of TS dicot stem (A) and TS dicot root (B).

- (a) With reference to Fig. 7.1A, identify the vascular tissues and associated cell layers in the dicot stem, stating the relative positions of xylem, phloem, cambium, and sclerenchyma fibres. [2]
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- (b) With reference to Fig. 7.1B, describe the arrangement of xylem, phloem, endodermis, and pericycle in the dicot root, contrasting this with the stem. [2]
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- (c) Explain how the mechanical arrangements of vascular bundles in the stem (peripheral ring) and stele in the root (central core) provide appropriate support against physical forces experienced by each organ. [2]
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2 9700/21/O/N/22/Q4 – Microscopic Anatomy of a Dicot Leaf and Midrib Vascular Bundle

[6 Marks] Syllabus 7.1

The internal anatomy of a dorsiventral dicot leaf reflects a precise division of labour between light harvesting, gas exchange, and vascular transport. Fig. 7.2 shows a transverse section through the lamina and midrib of a dicot leaf.

TS Dicot Leaf (Lamina & Midrib Vascular Bundle)

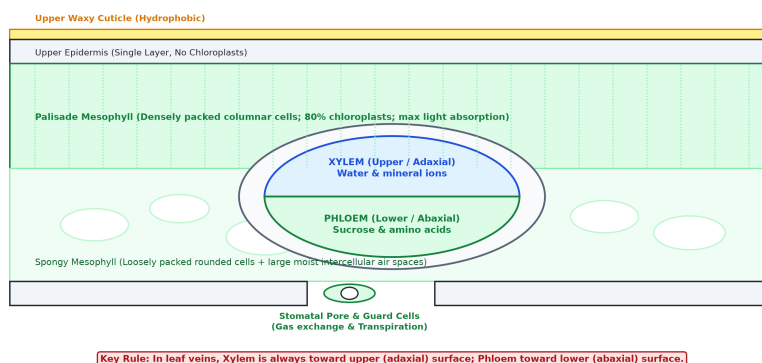


Fig. 7.2: TS plan diagram of a dicot leaf showing lamina layers and midrib vascular bundle.

- (a) Identify the positions of xylem and phloem in the leaf midrib vascular bundle in Fig. 7.2 and explain the developmental rationale for their spatial orientation relative to the stem. [2]
- (b) Compare the cellular adaptations of the palisade mesophyll with the spongy mesophyll in relation to their primary physiological functions. [2]
- (c) Describe how water moves from the leaf xylem to the atmosphere via mesophyll cells and stomata, stating the physical processes involved. [2]

3 9700/22/F/M/22/Q3 – Structural Adaptations of Xylem Vessel Elements for Water Transport

[6 Marks] Syllabus 7.1

Xylem vessel elements are highly specialised cells that undergo programmed cell death to form continuous conduits. Fig. 7.3 illustrates a longitudinal section through a xylem vessel and a transverse section showing secondary wall lignification.

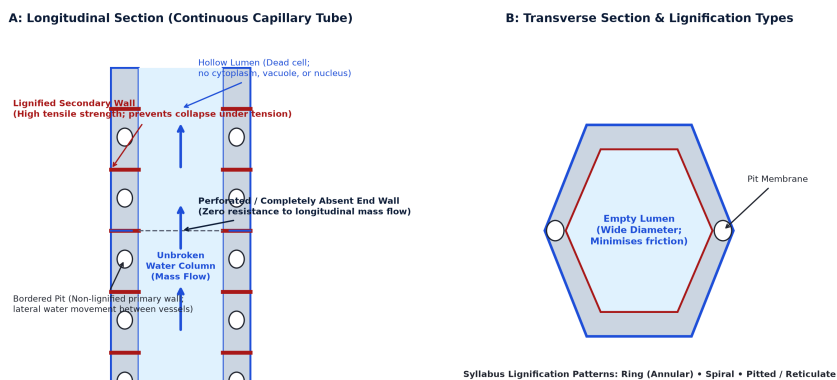


Fig. 7.3: Longitudinal section (A) and transverse section with wall thickenings (B) of xylem vessel elements.

- (a) With reference to Fig. 7.3A, describe two structural modifications that occur during xylem vessel differentiation that allow unimpeded mass flow of water. [2]
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- (b) Explain the mechanical role of lignin in xylem vessel walls and why this property is essential when the rate of transpiration is high. [2]
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- (c) State the structure and function of bordered pits in the lateral walls of adjacent xylem vessels, explaining their role in maintaining xylem function during cavitation. [2]
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9700/22/M/J/22/Q3 – Architecture and Interdependence of Phloem Sieve Tubes and Companion Cells

[6 Marks] Syllabus 7.1

Phloem tissue is composed of sieve tube elements and companion cells derived from the same parent meristematic cell. Fig. 7.4 shows the ultrastructure of a sieve tube element and an adjacent companion cell.

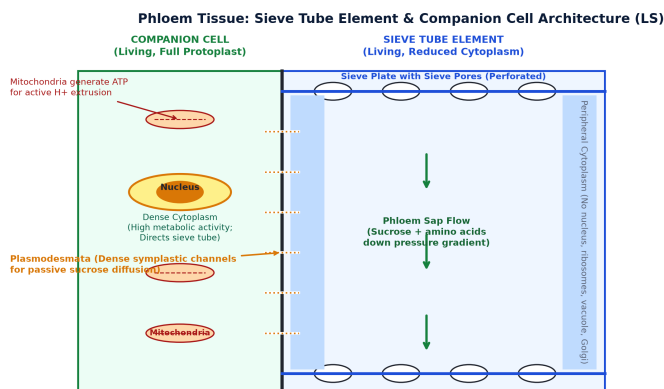


Fig. 7.4: Longitudinal ultrastructure of phloem sieve tube element and metabolic companion cell.

- (a) Describe the structural features of a mature sieve tube element shown in Fig. 7.4 that reduce resistance to the longitudinal flow of phloem sap. [2]
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- (b) Explain why sieve tube elements require companion cells to survive, identifying two ultrastructural features of companion cells that reflect their high metabolic activity. [2]
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- (c) Identify the microscopic channels linking companion cells to sieve tube elements in Fig. 7.4 and explain their functional importance in translocation. [2]
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5 9700/21/M/J/21/Q4 – Apoplast and Symplast Pathways and the Endodermal Casparian Strip

[6 Marks] Syllabus 7.2

Water absorbed by root hairs crosses the cortical parenchyma before entering the vascular cylinder. Fig. 7.5 illustrates the cellular pathways of water movement and the checkpoint established by the endodermis.

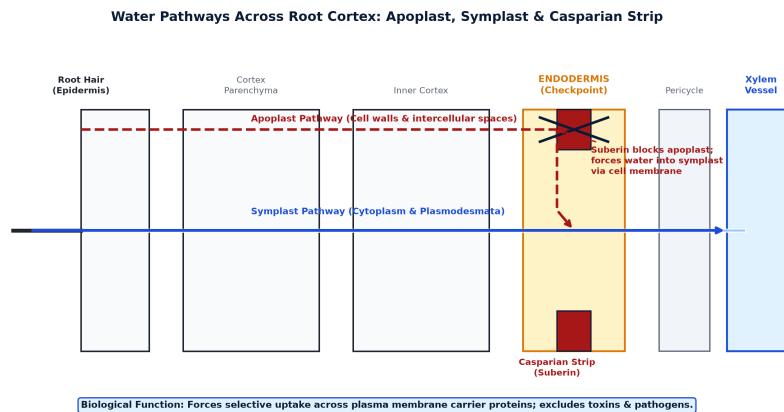


Fig. 7.5: Cellular pathways of water across root cortex and the suberised Casparian strip checkpoint.

- (a) With reference to Fig. 7.5, contrast the apoplast pathway with the symplast pathway in terms of the cellular structures traversed and physical resistance to water movement. [2]
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- (b) Explain the chemical composition of the Casparian strip and describe its effect on water and dissolved mineral ions arriving at the endodermis. [2]
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- (c) Discuss the physiological advantage to the plant of forcing water from the apoplast into the symplast at the endodermis. [2]
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6 9700/22/O/N/21/Q3 – Biophysical Mechanics of Cohesion–Tension Theory in Xylem Conduits

[6 Marks] Syllabus 7.2

The ascent of sap in tall trees occurs against gravity without a metabolic pump in the vascular tissue. Fig. 7.6 illustrates the generation of transpiration pull at the leaf mesophyll and the cohesive water column in xylem.

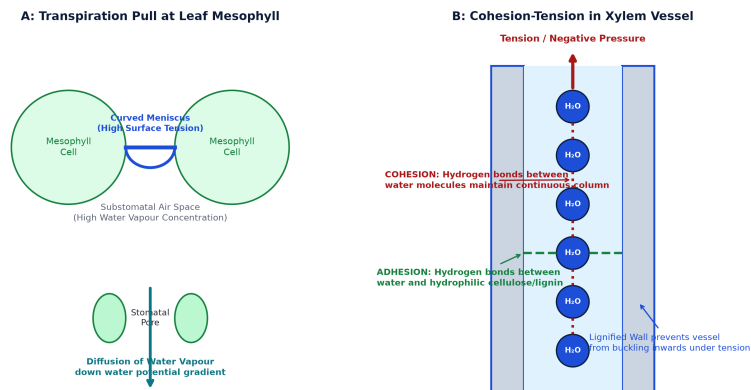


Fig. 7.6: Transpiration pull generated at mesophyll cell walls (A) and cohesion–tension in xylem vessel (B).

- (a) Explain how evaporation of water from mesophyll cell walls creates a negative hydrostatic pressure (tension) in leaf xylem. [2]
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- (b) Define cohesion and adhesion in the context of xylem transport, explaining the role of hydrogen bonding in each property. [2]
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- (c) Tree trunks exhibit a measurable diurnal fluctuation in diameter, contracting slightly during the middle of the day and expanding at night. Explain this phenomenon using the cohesion–tension theory. [2]
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7 9700/22/M/J/21/Q2 – Experimental Operation and Methodological Precautions of a Potometer

[6 Marks] Syllabus 7.2

A bubble potometer is widely employed in plant physiology laboratories to investigate rates of water movement. Fig. 7.7 illustrates the design of a calibrated capillary potometer.

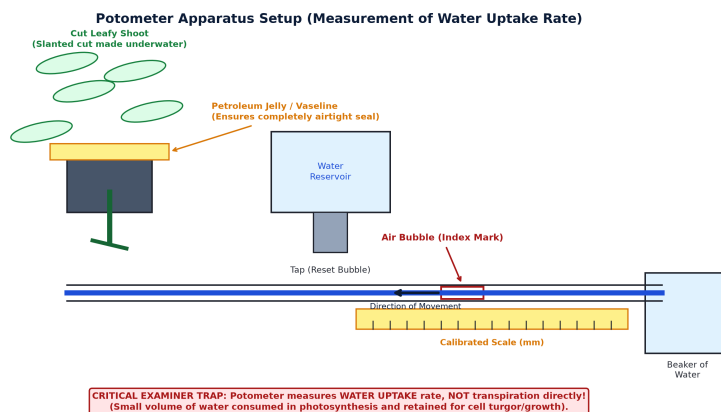


Fig. 7.7: Bubble potometer apparatus for measuring water uptake rate in a leafy shoot.

- (a) Describe two critical procedural steps that must be carried out when assembling the potometer apparatus to ensure valid and reliable measurements. [2]
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- (b) Outline the measurements required to calculate the absolute rate of water uptake in $\text{cm}^3 \text{min}^{-1}$ using the apparatus in Fig. 7.7. [2]
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- (c) Explain why the rate of water uptake measured by a potometer is NOT precisely equal to the true rate of transpiration. [2]
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8 9700/21/O/N/20/Q4 – Quantitative Evaluation of Environmental Factors Affecting Transpiration

[6 Marks] Syllabus 7.2

The rate of transpiration is influenced by environmental variables that alter the water potential gradient between the leaf and atmosphere. Fig. 7.8 shows experimental curves for transpiration rate against wind speed, humidity, light intensity, and temperature.

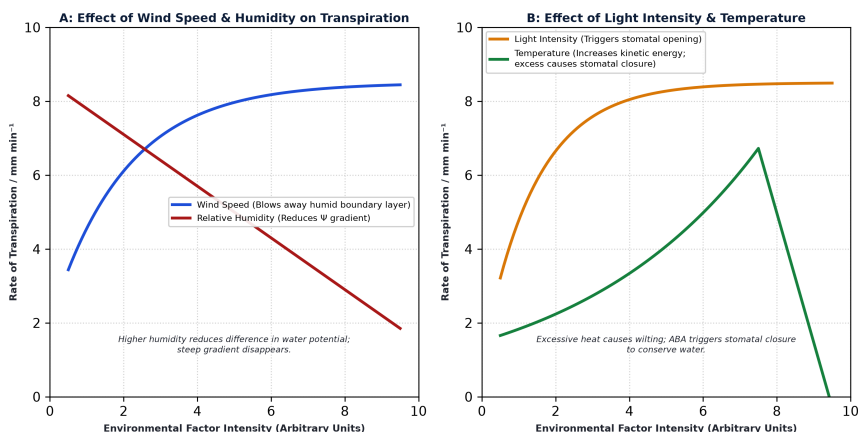


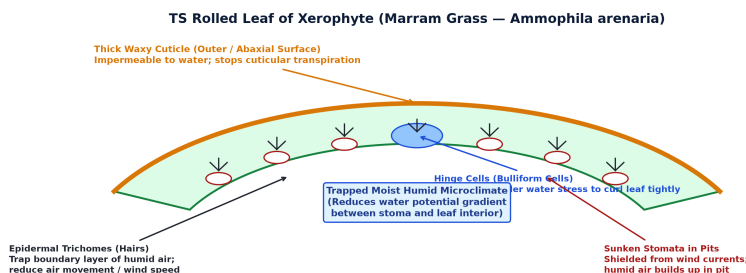
Fig. 7.8: Influence of wind speed and relative humidity (A) and light intensity and temperature (B) on transpiration rate.

- (a) With reference to Fig. 7.8A, explain the physiological mechanism by which increasing wind speed initially increases transpiration rate, and why the curve plateaus at high wind speeds. [2]
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- (b) With reference to Fig. 7.8A, explain why increasing relative humidity causes a steep decline in the rate of transpiration. [2]
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- (c) With reference to Fig. 7.8B, explain the shape of the temperature curve, accounting for the sharp decrease observed at extreme temperatures. [2]
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9 9700/22/M/J/20/Q3 – Morphological and Physiological Adaptations of Xerophytic Leaves

[6 Marks] Syllabus 7.2

Xerophytes inhabit environments where liquid water availability is severely restricted. Fig. 7.9 shows a transverse section through the rolled leaf of Marram grass (*Ammophila arenaria*).



Key Principle: Xerophytic adaptations work by increasing diffusion distance and reducing the water potential gradient.

Fig. 7.9: TS rolled leaf of Marram grass (*Ammophila arenaria*) showing xeromorphic adaptations.

- (a) Identify three distinct anatomical adaptations labelled in Fig. 7.9 that restrict water loss in Marram grass. [2]
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- (b) Explain how epidermal trichomes (hairs) and sunken stomata in grooves operate biophysically to reduce transpiration. [2]
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- (c) Describe the role of hinge cells (bulliform cells) in the rolling and unrolling mechanism of the leaf in response to atmospheric humidity. [2]
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10 9700/21/M/J/19/Q4 – Chemiosmotic Mechanism of Sucrose Loading into Phloem Companion Cells

[6 Marks] Syllabus 7.2

Sucrose synthesised in photosynthesising mesophyll source cells is actively loaded into the phloem apoplast. Fig. 7.10 illustrates the molecular mechanism operating in the companion cell plasma membrane.

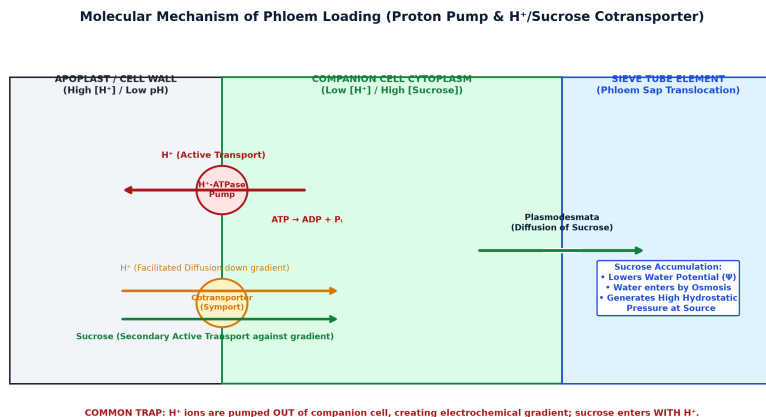


Fig. 7.10: Active loading of sucrose via H⁺-ATPase proton pump and H⁺/sucrose cotransporter.

- (a) With reference to Fig. 7.10, describe the primary active transport step carried out by the proton pump in the companion cell membrane. [2]
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- (b) Explain how the electrochemical proton gradient drives the secondary active transport of sucrose into the companion cell via the cotransporter protein. [2]
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- (c) Predict the effect on sucrose translocation if companion cells are treated with a metabolic poison such as cyanide or dinitrophenol (DNP). [2]
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11 9700/22/O/N/19/Q3 – The Mass Flow Hypothesis of Translocation Down Hydrostatic Pressure Gradients [6 Marks] Syllabus 7.2

The mass flow (pressure–flow) hypothesis explains how organic assimilates are translocated over long distances in sieve tube elements. Fig. 7.11 shows the hydrostatic pressure model operating between source and sink.

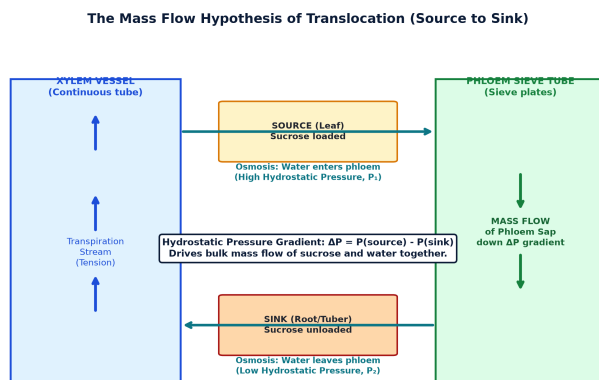


Fig. 7.11: The Münch mass flow hypothesis model linking source, sink, phloem, and xylem.

- (a) With reference to Fig. 7.11, explain why active loading of sucrose at the source generates a high hydrostatic pressure inside sieve tube elements. [2]
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- (b) Describe the events occurring at the sink tissue that generate a low hydrostatic pressure inside sieve tube elements. [2]
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- (c) State the physical principle that causes phloem sap to move from source to sink and state two experimental observations that support the mass flow hypothesis. [2]
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12 9700/22/M/J/18/Q4 – Classical Aphid Stylet and Radiotracer Evidence for Phloem Transport

[6 Marks] Syllabus 7.2

Plant physiologists utilised aphids (e.g. *Aphis fabae*) to sample pristine phloem sap without triggering callose sealing. Fig. 7.12 illustrates an aphid stylet penetrating a sieve tube and a radioactive ^{14}C pulse-chase translocation profile.

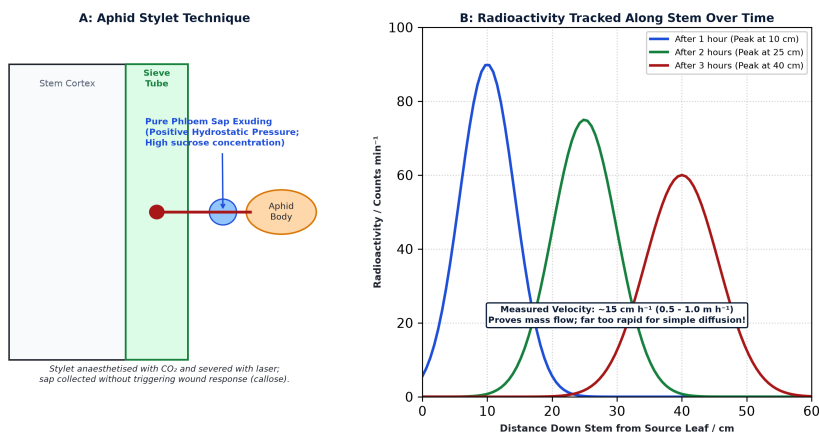


Fig. 7.12: Aphid stylet sampling (A) and ^{14}C -sucrose velocity profile along stem over time (B).

- (a) Explain why severing the aphid body from its inserted stylet allows pure phloem sap to continue exuding from the cut stump for many hours. [2]
- (b) With reference to Fig. 7.12B, calculate the translocation velocity of ^{14}C -labelled sucrose in cm h^{-1} between 1 hour and 3 hours, and evaluate whether this transport could occur by simple diffusion. [2]
- (c) Chemical analysis of the exuded sap reveals sucrose, amino acids, ATP, and potassium ions, but no glucose or starch. Explain the physiological significance of translocating sucrose rather than glucose. [2]

13 9700/21/O/N/18/Q4 – Classical Girdling Experiments Investigating Translocation Pathways

[6 Marks] Syllabus 7.1

In the 17th century, Marcello Malpighi pioneered girdling (ringing) experiments on woody tree branches. Fig. 7.13 shows the experimental appearance of a woody stem immediately after removing a ring of bark and several weeks later.

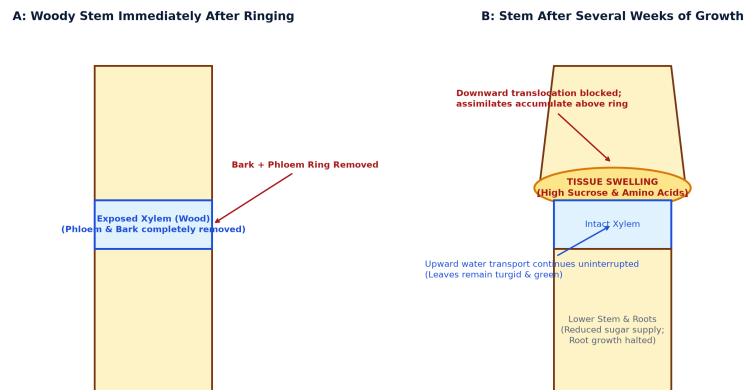


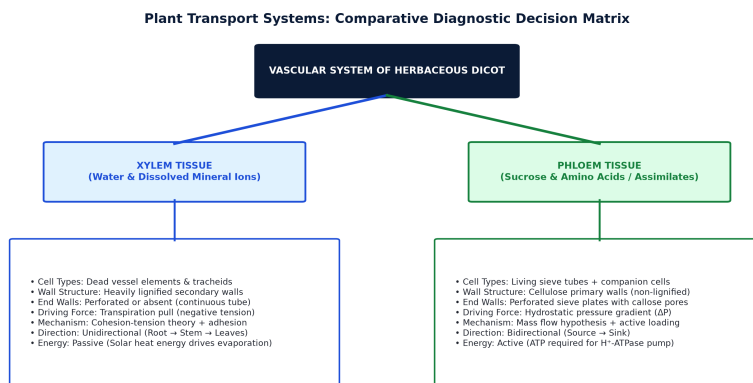
Fig. 7.13: Stem girdling experiment immediately after ringing (A) and after several weeks of growth (B).

- (a) With reference to Fig. 7.13A, identify the specific tissues removed when a complete ring of bark is excised from a woody dicot stem. [2]
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- (b) Explain why tissue swelling occurs exclusively ABOVE the ring in Fig. 7.13B, describing the biochemical contents of the swollen zone. [2]
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- (c) Explain why leaves on the upper branches remain green and turgid for weeks following girdling, but the entire tree eventually dies if the girdle is left unsealed. [2]
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14 9700/22/F/M/18/Q3 – Comprehensive Comparative Analysis of Xylem and Phloem Transport

[6 Marks] Syllabus 7.1

Vascular plants possess two distinct, specialised long-distance transport networks. Fig. 7.14 presents an analytical classification tree comparing the structural, functional, and energetic properties of xylem and phloem.



Cambridge Marking Criterion: Always link tissue anatomy directly to biophysical transport physics!

Fig. 7.14: Decision matrix and comparative properties of xylem vs phloem transport systems.

- (a) With reference to Fig. 7.14, complete a rigorous structural comparison between a mature xylem vessel element and a mature phloem sieve tube element. [2]

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- (b) Contrast the physical driving forces responsible for liquid movement in xylem with those operating in phloem, stating the pressure regimes (positive vs negative) in each conduit. [2]

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- (c) Compare the metabolic energy requirements of xylem transport with phloem transport, explaining the ultimate energy source powering each system. [2]

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15 9700/22/M/J/17/Q3 – Water and Mineral Ion Uptake by Root Hair Cells

[6 Marks] Syllabus 7.2

Root hair cells are specialised epidermal trichoblasts that mediate the primary absorption of water and dissolved inorganic ions from the soil solution.

- (a) Describe how the morphology of root hair cells adapts them for rapid absorption of water and inorganic mineral ions. [2]

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- (b) Explain why the uptake of nitrate (NO_3^-) and potassium (K^+) ions into root hair cells requires metabolic energy (ATP), whereas water absorption is passive. [2]

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- (c) Explain how the active accumulation of mineral ions inside the root hair vacuole indirectly facilitates the uptake of water from soil. [2]

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16 9700/21/O/N/16/Q3 – Biophysical Mechanism of Stomatal Opening and Closing in Guard Cells

[6 Marks] Syllabus 7.2

Stomata function as dynamic valves that regulate the compromise between photosynthetic carbon dioxide acquisition and transpirational water conservation.

- (a) Describe the ultrastructural features of guard cell walls that cause them to bend and open a central pore when they become turgid. [2]

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- (b) Explain the sequence of ion movements and osmotic events that leads to stomatal opening in response to dawn light. [2]

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- (c) Explain how the plant hormone abscisic acid (ABA) triggers stomatal closure during severe soil drought stress. [2]

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17 9700/22/M/J/16/Q4 – Seasonal Dynamics of Sources and Sinks in Perennial Angiosperms

[6 Marks] Syllabus 7.2

In angiosperms, organs that act as sources or sinks for organic assimilates are not permanent, but undergo reversible functional transitions across seasons.

- (a) Define the terms 'source' and 'sink' in relation to phloem translocation, providing two distinct examples of each in a potato plant during mid-summer. [2]

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- (b) Explain how a storage organ such as a potato tuber undergoes a complete reversal from sink to source between autumn and spring. [2]

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- (c) Explain why young developing leaves initially function as sinks before transitioning into sources as they mature. [2]

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18 9700/21/M/J/15/Q4 – Root Pressure Generation and Guttation Under Low Transpiration Conditions

[6 Marks] Syllabus 7.2

Under environmental conditions where transpiration is virtually zero, plants can generate positive hydrostatic root pressure.

- (a) Describe how endodermal cells actively generate positive hydrostatic root pressure in root xylem vessels. [2]

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- (b) Explain why root pressure alone is insufficient to account for the ascent of water to the tops of tall trees such as Sequoia sempervirens. [2]

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- (c) Define guttation, stating the specialised structures through which guttation droplets exude, and contrast guttation with morning dew. [2]

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19 [Mentora Original A* Extension] – Quantitative Hydraulic Conductance and Cavitation Vulnerability [6 Marks] Syllabus 7.2

When xylem vessels experience severe negative tensions during drought, liquid water can vaporise explosively, forming an air bubble (embolism) in a process called cavitation.

- (a) Explain how cavitation occurs in xylem vessels and describe how air seeding occurs through bordered pit membranes from an embolised vessel to an adjacent water-filled vessel. [2]

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- (b) A plant physiologist measures a Cavitation Vulnerability Curve showing Percent Loss of Hydraulic Conductance (PLC) against xylem water potential (Psi). State the expected mathematical relationship and define P50. [2]

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- (c) Discuss two anatomical strategies evolved by drought-tolerant woody plants that minimise the spread of embolisms across the xylem network. [2]

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20 [Mentora Original A* Extension] – Biophysics of Stomatal Conductance and Fick's Law of Diffusion [6 Marks] Syllabus 7.2

Transpirational flux from leaves is governed by Fick's First Law of Diffusion: $J = -D * (\Delta C / \Delta x)$, where total resistance is the sum of stomatal resistance and boundary layer resistance.

- (a) Using Fick's Law of Diffusion, explain why splitting a single large stomatal pore into multiple small pores of equivalent total area drastically increases the total transpirational diffusion flux. [2]

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- (b) Distinguish between stomatal conductance (g_s) and boundary layer conductance (g_b), explaining which parameter is under direct physiological control by the plant. [2]

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- (c) Define Transpiration Water Use Efficiency (WUE) at the leaf level, explaining why C4 plants achieve significantly higher WUE than C3 plants under high temperatures. [2]

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21 9700/22/M/J/23/Q4 – Comparative Morphology of Xylem Vessels and Tracheids

[4 Marks] Syllabus 7.1

Angiosperms possess both xylem vessel elements and tracheids, whereas gymnosperms rely almost exclusively on tracheids for water conduction.

- (a) Contrast the morphology and end-wall structure of a tracheid with that of a xylem vessel element. [2]

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- (b) Explain the functional trade-off between hydraulic efficiency and safety against cavitation in vessels compared to tracheids. [2]

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22 9700/21/O/N/22/Q3 – Role of Callose and P-protein in Phloem Wound Responses

[4 Marks] Syllabus 7.1

When phloem sieve tubes are punctured or damaged by herbivores, plants execute an immediate sealing reaction to prevent catastrophic loss of nutrient-rich sap.

- (a) Describe the rapid sealing response of phloem sieve tube elements upon physical wounding or sudden drop in turgor pressure. [2]

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- (b) Explain the chemical nature of callose and describe how its deposition is regulated during normal seasonal dormancy. [2]

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23 9700/22/M/J/22/Q4 – Pathways of Water Movement from Leaf Xylem to Stomatal Pores

[4 Marks] Syllabus 7.2

Water exiting the terminal vein endings in a leaf must traverse several tissue layers before evaporating into the atmosphere.

- (a) Trace the exact route taken by water molecules from the lumen of a leaf vein xylem vessel to the outer atmosphere. [2]

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- (b) Explain why the intercellular air spaces of the spongy mesophyll remain saturated with water vapour (relative humidity ~99%). [2]

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24 9700/21/M/J/22/Q3 – Anatomical Adaptations of Hydrophytic Leaves (Nymphaea)

[4 Marks] Syllabus 7.1

Water lilies (*Nymphaea*) inhabit aquatic environments and exhibit leaf specialisations that contrast markedly with terrestrial xerophytes.

- (a) Describe two structural adaptations of floating *Nymphaea* leaves related to stomatal distribution and internal gas exchange. [2]

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- (b) Explain why xylem tissue is greatly reduced in hydrophytes compared to terrestrial plants. [2]

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25 9700/22/F/M/21/Q2 – Apoplastic vs Symplastic Mechanisms of Phloem Loading

[4 Marks] Syllabus 7.2

While many herbaceous crop species load sucrose via the apoplastic route using proton pumps, some woody species employ symplastic loading.

- (a) Distinguish between apoplastic and symplastic phloem loading in terms of membrane transport proteins and plasmodesmata involvement. [2]

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- (b) Explain the polymer trapping hypothesis that enables symplastic loaders (e.g. *Cucurbita*) to accumulate sugars against a concentration gradient. [2]

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26 9700/21/O/N/21/Q4 – Physical Boundary Layer and Transpiration Dynamics

[4 Marks] Syllabus 7.2

The boundary layer is a thin layer of still, unstirred air that adheres to the outer surface of a leaf lamina.

- (a) Explain how boundary layer thickness affects the total resistance to water vapour diffusion out of a leaf. [2]

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- (b) Describe how leaf size and surface texture (e.g. hairs, waxy ridges) influence the boundary layer thickness in windy conditions. [2]

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27 9700/22/M/J/20/Q4 – Differentiation and Wall Patterns of Protoxylem and Metaxylem

[4 Marks] Syllabus 7.1

During primary growth in young stems and roots, xylem elements differentiate sequentially as protoxylem and metaxylem.

- (a) Contrast the wall lignification patterns and lumen diameters of protoxylem with metaxylem. [2]

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- (b) Explain why protoxylem possesses annular or spiral lignin rings rather than complete pitted secondary walls. [2]

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28 9700/22/F/M/20/Q3 – Suberin Deposition and Passage Cells in the Endodermis

[4 Marks] Syllabus 7.1

As roots mature, endodermal cells deposit secondary and tertiary suberin and lignin lamellae.

- (a) Describe the developmental stages of endodermal wall thickening from primary Casparian strip to tertiary U-shaped thickenings. [2]

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- (b) State the location and function of 'passage cells' in older endodermal rings. [2]

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29 9700/21/M/J/19/Q3 – Nitrogen Assimilation and Translocation in Xylem vs Phloem

[4 Marks] Syllabus 7.2

Nitrogen absorbed by roots as nitrate or ammonium is converted into organic compounds before long-distance transport.

- (a) Compare the forms in which nitrogen compounds are translocated in xylem sap with those in phloem sap. [2]

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- (b) Explain how amino acids are loaded into phloem companion cells at source tissues. [2]

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30 9700/22/M/J/19/Q4 – Quantitative Relationship Between Solute Potential and Pressure Potential

[4 Marks] Syllabus 7.2

The water potential equation for plant cells is formulated as: $\Psi = \Psi_s + \Psi_p$.

- (a) Define solute potential (Ψ_s) and pressure potential (Ψ_p), stating whether each value is typically positive, negative, or zero in a living turgid mesophyll cell. [2]

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- (b) Calculate the overall water potential (Ψ) of a plant cell with $\Psi_s = -850$ kPa and $\Psi_p = +350$ kPa, predicting the direction of net water movement if placed in a sucrose solution of $\Psi = -600$ kPa. [2]

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31 9700/22/O/N/18/Q3 – Acoustic Detection of Xylem Cavitation Events

[4 Marks] Syllabus 7.2

When xylem vessels cavitate under tension, ultrasonic acoustic emissions (clicking sounds) can be detected with sensitive transducers.

- (a) Explain the physical origin of ultrasonic acoustic emissions during xylem cavitation. [2]

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- (b) Explain why the frequency of acoustic emissions increases dramatically during peak afternoon hours in drought-stressed plants. [2]

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32 9700/21/M/J/18/Q4 – Ultrastructure of Phloem Transfer Cells (Modified Companion Cells)

[4 Marks] Syllabus 7.1

In many angiosperm leaves, companion cells differentiate into specialised 'transfer cells' with labyrinthine wall ingrowths.

- (a) Describe the structural appearance of cell wall ingrowths in transfer cells and explain their functional benefit. [2]

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- (b) Explain how the proliferation of wall ingrowths directly enhances the rate of secondary active sucrose cotransport. [2]

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33 9700/22/F/M/17/Q3 – Effect of Low Temperatures on Phloem Translocation Velocity

[4 Marks] Syllabus 7.2

Cooling a localized collar of plant stem to 0–2 deg C severely retards the translocation of radioactive tracers.

- (a) Explain why chilling a section of stem slows phloem translocation, distinguishing between enzymatic loading and sap viscosity. [2]

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- (b) Explain why cooling stem xylem does not have a comparable immediate inhibitory effect on water movement. [2]

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34 9700/22/M/J/17/Q4 – Comparative Stomatal Densities in Mesophytes and Xerophytes

[4 Marks] Syllabus 7.1

Stomatal density (number of stomata per unit leaf area) varies widely across ecological plant types.

- (a) Outline a microscopic method to determine the stomatal density of an epidermis using a clear nail varnish peel and graticule. [2]

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- (b) Explain why many xerophytes have stomata restricted exclusively to the lower (abaxial) epidermis or inside grooved pits. [2]

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35 9700/21/O/N/16/Q4 – Distinction Between Transpiration and Physical Evaporation

[4 Marks] Syllabus 7.2

While transpiration involves evaporation, it is subject to physiological regulation unique to living plant tissues.

- (a) State two essential differences between transpiration from a plant canopy and evaporation from an open lake surface. [2]

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- (b) Explain how cuticular transpiration differs from stomatal transpiration in its proportion and physiological control. [2]

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36 9700/22/M/J/15/Q3 – Embolism Formation and Mechanisms of Xylem Refilling

[4 Marks] Syllabus 7.2

When a xylem vessel embolises, the continuous water column is severed, rendering that conduit non-functional.

- (a) Explain why air bubbles in xylem vessels cannot be pulled upward by transpiration pull. [2]

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- (b) Describe how positive root pressure at night or active solute pumping by xylem parenchyma cells can dissolve and refill embolised vessels. [2]

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37 9700/21/M/J/15/Q2 – Histological Comparison of Collenchyma and Sclerenchyma Support Tissues

[4 Marks] Syllabus 7.1

Primary plant stems rely on collenchyma and sclerenchyma for structural support alongside turgid parenchyma.

- (a) Compare the cell wall composition and vitality (living vs dead) of collenchyma cells with sclerenchyma fibres. [2]

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- (b) Explain how the mechanical properties of collenchyma allow it to support young growing stems without restricting growth. [2]

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38 9700/22/O/N/15/Q4 – Pathways of Sucrose Unloading in Sink Tissues (Meristems vs Storage Organs)

[4 Marks] Syllabus 7.2

Once sucrose reaches a sink organ, it must exit the phloem to be utilised in metabolism or polymerised for storage.

- (a) Describe the symplastic pathway of sucrose unloading into rapidly growing apical meristems. [2]

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- (b) Describe the apoplastic pathway of sucrose unloading in storage tissues, identifying the role of cell wall invertase. [2]

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39 [Mentora Original A* Extension] – Hydrodynamics of Phloem Sap: Poiseuille's Law and Sieve Plate Resistance [4 Marks] Syllabus 7.2

Phloem sap is a viscous solution (~10–25% w/v sucrose). Sap flux through a cylindrical lumen of radius r follows Poiseuille's law: $Q = (\pi \cdot r^4 \cdot \Delta P) / (8 \cdot \eta \cdot L)$.

- (a) Explain why small increases in the radius of sieve tube elements have an overwhelmingly large effect on volume flow rate (Q). [2]

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- (b) Explain why sieve plates represent over 50% of the total hydrodynamic resistance in a phloem conduit despite occupying less than 1% of its total length. [2]

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40 [Mentora Original A* Extension] – Maximum Theoretical Height of Trees and Tensile Strength of Water [4 Marks] Syllabus 7.2

The maximum height of tall trees (e.g. Coastal Redwood, ~115 m) is physically constrained by gravity, frictional resistance, and water tensile strength.

- (a) Calculate the minimum hydrostatic tension required at the top of a 100 m tree, assuming a gravitational gradient of 0.01 MPa m^{-1} and frictional resistance equal to gravitational gradient. [2]

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- (b) Explain why coastal fog absorption through needles (foliar water uptake) is critical for sustaining water potential in the upper crowns of the world's tallest redwoods. [2]

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41 9700/22/M/J/23/Q1 – Precise Definition of Transpiration [2 Marks] Syllabus 7.2

Transpiration is a fundamental consequence of gas exchange in terrestrial autotrophs.

- (a) Provide the precise Cambridge definition of transpiration, stating the two distinct physical processes involved. [2]

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42 9700/21/O/N/22/Q1 – Precise Definition of Translocation [2 Marks] Syllabus 7.2

Translocation mediates assimilate distribution throughout vascular plants.

- (a) Define translocation, identifying the transport tissue, the primary chemical substances transported, and directionality. [2]

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43 9700/22/F/M/22/Q1 – Chemical Nature and Role of the Casparian Strip

[2 Marks] Syllabus 7.2

The endodermis contains a specialised hydrophobic barrier in primary roots.

- (a) State the precise chemical composition of the Casparian strip and explain its exact biological effect on apoplastic water flow. [2]

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44 9700/22/M/J/22/Q1 – Apoplast and Symplast Definitions

[2 Marks] Syllabus 7.2

Water moves through plant tissues along interconnected cellular compartments.

- (a) Define the apoplast and symplast pathways in plant tissues. [2]

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45 9700/21/M/J/21/Q1 – Mechanical and Physical Properties of Lignin

[2 Marks] Syllabus 7.1

Lignification of xylem walls was a pivotal adaptation in terrestrial plant evolution.

- (a) State two essential physical properties of lignin that adapt xylem vessels for their dual role in transport and support. [2]

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46 9700/22/O/N/21/Q1 – Bordered Pits in Xylem Conduits

[2 Marks] Syllabus 7.1

Bordered pits are prominent features of xylem vessel element walls.

- (a) Describe the structure of a bordered pit and state its primary functional role in lateral water movement. [2]

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47 9700/22/F/M/21/Q1 – Functional Specialisation of Companion Cells

[2 Marks] Syllabus 7.1

Companion cells are metabolically coupled to sieve tube elements.

- (a) State two metabolic roles performed by companion cells that support adjacent sieve tube elements. [2]

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48 9700/21/O/N/20/Q1 – Xerophytic Leaf Diagnostics

[2 Marks] Syllabus 7.2

Xerophytes exhibit distinctive leaf modifications to survive arid conditions.

- (a) Name one specific xerophytic plant species and state two diagnostic xeromorphic leaf adaptations it possesses. [2]

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49 9700/22/M/J/20/Q1 – Ultrastructure of Sieve Plates

[2 Marks] Syllabus 7.1

Sieve plates demarcate the junction between successive sieve tube elements.

- (a)** Describe the structure of a sieve plate and explain how its presence influences longitudinal resistance to sap flow. **[2]**

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50 9700/21/M/J/19/Q1 – Scientific Principle of Potometer Measurements

[2 Marks] Syllabus 7.2

Potometers are standard apparatus in botanical transport investigations.

- (a)** State precisely what physical quantity a potometer measures and state the core assumption made when inferring transpiration rate. **[2]**

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SECTION D: 10 HIGH-FREQUENCY EXAMINER FAQs & CRITICAL REVISION PITFALLS

Essential Cambridge International AS Level Biology (9700) Mark-Scheme Traps, Core Misconceptions, and Model Answers

PHLOEM TRANSLOCATION • CHEMIOSMOTIC LOADING

FAQ 1: How Does Active Sucrose Loading Occur in Companion Cells and What Is the Exact Role of the Proton Gradient?

Cambridge Examiner Warning: Candidates frequently lose marks by confusing the direction of proton pumping (pumping H^+ OUT into the cell wall/apoplast, NOT into the cell) or stating that sucrose is directly transported by an ATP-driven pump.

- H^+ -ATPase proton pumps actively hydrolyse ATP ($ATP \rightarrow ADP + Pi$) to pump protons (H^+ ions) OUT of companion cell cytoplasm into the cell wall/apoplast.
- This establishes a steep electrochemical proton gradient (high $[H^+]$ and positive electrical charge in apoplast; low $[H^+]$ in cytoplasm).
- Protons diffuse passively back down their concentration gradient through H^+ /sucrose cotransporter proteins (secondary active symport).
- Energy released by protons moving down gradient drives simultaneous cotransport of sucrose AGAINST its concentration gradient into companion cell.
- Sucrose then diffuses passively down concentration gradient into sieve tube elements through numerous plasmodesmata.

MASS FLOW HYPOTHESIS • TRANSLOCATION PHYSICS

FAQ 2: Explain the Complete Mass Flow Hypothesis: How Does a Hydrostatic Pressure Gradient Drive Sap from Source to Sink?

Cambridge Examiner Warning: Examiners penalise students who state that sucrose moves by diffusion down a concentration gradient in the phloem. Translocation is MASS FLOW driven by a HYDROSTATIC PRESSURE GRADIENT (ΔP).

- At Source (photosynthesising leaves): Active loading of sucrose into sieve tubes lowers water potential (Ψ) inside the phloem sap.
- Water enters sieve tubes from adjacent xylem vessels by osmosis down a water potential gradient.
- Influx of water into rigid, non-expanding cell walls generates a high hydrostatic pressure (turgor pressure, P_{source}).
- At Sink (roots, tubers, fruits): Sucrose is unloaded for respiration or converted to insoluble starch, raising water potential inside phloem.
- Water exits sieve tubes by osmosis into surrounding tissues and xylem, creating a low hydrostatic pressure (P_{sink}).
- Hydrostatic pressure gradient ($\Delta P = P_{\text{source}} - P_{\text{sink}}$) drives bulk mass flow of water and dissolved sucrose together from source to sink.

ROOT WATER PATHWAYS • ENDODERMAL CHECKPOINT

FAQ 3: What Is the Casparian Strip and Why Does It Force Water from the Apoplast to the Symplast Pathway at the Endodermis?

Cambridge Examiner Warning: Candidates commonly state that the Casparian strip 'stops water completely' (which would kill the plant) or incorrectly locate it in the epidermis or cortex rather than the endodermis.

- Across the root cortex, water moves predominantly via the apoplast pathway (through cellulose cell walls and intercellular spaces) with minimal resistance.
- The Casparian strip is a band of waterproof, waxy suberin deposited in the radial and transverse walls of endodermal cells.
- Suberin is impermeable to water, completely blocking further apoplastic flow at the endodermis.
- Water and dissolved mineral ions are forced to cross the selectively permeable plasma membrane into the living symplast pathway (cytoplasm).
- Significance: Allows plant to actively regulate ion uptake via carrier proteins, exclude soil toxins/pathogens, and prevent ion back-leakage from xylem.

COHESION-TENSION THEORY • XYLEM MECHANICS

FAQ 4: How Does Transpiration Pull Drive the Ascent of Water in Xylem Vessels According to Cohesion-Tension Theory?

Cambridge Examiner Warning: Stating that water is 'pushed up by positive pressure' or 'pulled up by osmosis' scores 0 marks. Examiners insist on transpiration pull (tension / negative pressure) and cohesion due to hydrogen bonds.

- Transpiration pull: Water evaporates from wet mesophyll cell walls into leaf air spaces and diffuses through stomata; water menisci in wall microfibrils retreat, generating tension (negative hydrostatic pressure).
- Cohesion: Polar water molecules form hydrogen bonds with one another, holding the water column together as a continuous, unbroken thread with high tensile strength.
- Adhesion: Water molecules form hydrogen bonds with hydrophilic cellulose and lignin in xylem vessel walls, supporting column weight against gravity.
- As water evaporates at the leaves, the entire unbroken water column is pulled upward under tension through xylem from roots to foliage.

XYLEM ADAPTATIONS • ULTRASTRUCTURE & FUNCTION

FAQ 5: What Structural Adaptations Enable Xylem Vessel Elements to Function as High-Efficiency Capillary Conduits?

Cambridge Examiner Warning: Asserting that lignin's primary role is 'waterproofing to keep water in' misses the key mechanical function: preventing vessel collapse under extreme negative pressure / tension.

- Dead, hollow cells: Complete autolysis of protoplast (no cytoplasm, vacuole, or nucleus) leaves an open lumen, minimising frictional resistance to bulk flow.
- Absent or perforated end walls: End-to-end stacking of vessel elements forms continuous, uninterrupted capillary tubes of wide diameter.
- Lignified secondary walls: Secondary walls impregnated with lignin provide immense tensile strength, preventing walls from buckling inward under tension.
- Non-lignified bordered pits: Thin primary walls in lateral pits permit lateral water transfer between parallel vessels, bypassing embolisms/air locks.

PHLOEM TISSUE HISTOLOGY • CELL COMPARISONS

FAQ 6: Contrast the Structural and Metabolic Specialisations of Sieve Tube Elements and Companion Cells.

Cambridge Examiner Warning: Candidates often incorrectly claim that sieve tube elements are 'dead cells' (like xylem) or confuse which cell possesses the nucleus, mitochondria, and ribosomes.

- Sieve Tube Elements (Living, Reduced): Retain living plasma membrane but lose nucleus, central vacuole, ribosomes, and Golgi; peripheral cytoplasm lines walls around an open central lumen; end walls perforated as sieve plates with callose pores to facilitate mass flow.
- Companion Cells (Living, Highly Metabolic): Possess prominent nucleus, dense cytoplasm, and abundant mitochondria to synthesise ATP for H^+ -ATPase pumps.
- Metabolic coupling: Companion cells synthesise essential proteins and perform metabolic maintenance for adjacent sieve tube elements.
- Symplastic linkage: Joined by dense clusters of branched plasmodesmata, providing direct channels for sucrose, ATP, and signalling molecules.

XEROMORPHIC ANATOMY • WATER CONSERVATION

FAQ 7: How Do Sunken Stomata, Epidermal Trichomes, and Rolled Leaves Reduce Transpiration in Xerophytes?

Cambridge Examiner Warning: Vague answers like 'reduces water loss' earn no credit. Mark schemes strictly require the physical mechanism: trapping a boundary layer of humid air and reducing the water potential gradient.

- Sunken Stomata in Pits/Grooves: Stomata are sheltered from external wind; water vapour evaporating from pores accumulates in the pit, creating a localized humid microclimate.
- Epidermal Trichomes (Hairs): Interlocking hairs trap a stagnant boundary layer of moist, humid air next to the leaf surface, shielding it from wind currents.
- Rolled/Curled Leaves (e.g. Marram grass): Hinge cells lose turgor to roll leaf cylinder, enclosing all stomata inside an internal humid chamber.
- Common Mechanism: All three adaptations increase diffusion distance and reduce the steepness of the water potential gradient between substomatal cavity and outside air.

EXPERIMENTAL BOTANY • POTOMETER METHODOLOGY

FAQ 8: What Does a Potometer Actually Measure, What Are Critical Setup Precautions, and Why Is Uptake NOT Equal to Transpiration?

Cambridge Examiner Warning: Stating that a potometer 'measures transpiration rate directly' is an immediate examiner trap. A potometer measures WATER UPTAKE.

- Physical Quantity Measured: A potometer measures the rate of WATER UPTAKE by a leafy shoot, based on the assumption that uptake approximately equals transpiration under steady-state conditions.
- Critical Precautions: (1) Cut shoot underwater at a slanting angle to prevent air locks in xylem and increase surface area; (2) Assemble apparatus underwater and seal joints with petroleum jelly for an airtight system.
- Why Uptake Exceeds Transpiration: (1) Water is retained in vacuoles to maintain cell turgidity and allow cell elongation/growth; (2) Water is chemically consumed as a reactant in photosynthesis (photolysis in PSII); (3) Water used in metabolic hydrolysis.

PLANT HISTOLOGY • TISSUE PLAN DIAGRAMS

FAQ 9: How Do You Accurately Distinguish and Draw Vascular Tissues in Low-Power Plan Diagrams of Dicot Stem, Root, and Leaf?

Cambridge Examiner Warning: Drawing individual cells in low-power plan diagrams is forbidden in Cambridge exams and receives an immediate zero. Reversing xylem and phloem positions is also heavily penalised.

- Dicot Stem TS: Vascular bundles arranged in a peripheral ring surrounding wide central parenchyma pith; Xylem on inner side (facing pith), Phloem on outer side (facing cortex), separated by vascular cambium; sclerenchyma fibres cap phloem.
- Dicot Root TS: Vascular tissue forms a central stele/cylinder (no central pith); Xylem forms a central star/cross (tetrarch core); Phloem clusters located between arms of xylem star; bounded by pericycle and endodermis.
- Dicot Leaf Midrib TS: Vascular bundle oriented with Xylem on upper (adaxial) surface and Phloem on lower (abaxial) surface, reflecting leaf trace divergence.
- Plan Diagram Rules: Draw clean, unbroken lines indicating tissue boundaries only; do NOT draw individual cells; maintain correct tissue proportions.

WATER TRANSPORT PATHWAYS • CELLULAR PHYSICS

FAQ 10: Compare the Apoplastic, Symplastic, and Vacuolar Pathways of Water Movement Across Root Cortex Cells.

Cambridge Examiner Warning: Candidates confuse apoplast (cell walls) with symplast (cytoplasm and plasmodesmata) or incorrectly suggest that apoplastic water movement requires membrane transport proteins.

- Apoplast Pathway: Water moves entirely through porous cellulose cell walls and intercellular spaces by capillary action and mass flow; does not cross any plasma membranes; offers lowest resistance to water flow.
- Symplast Pathway: Water enters root hair cell by osmosis across plasma membrane; moves through continuous living cytoplasm from cell to cell via plasmodesmata down a water potential gradient; higher resistance than apoplast.
- Vacuolar Pathway: Water moves from vacuole to vacuole across cell walls, plasma membranes, and tonoplasts by osmosis; encounters highest resistance and represents a minor component of total water transport.
- Biological Junction: At the endodermis, the suberised Casparian strip halts apoplastic flow, forcing 100% of water into the symplast.

EXAMINER MARK SCHEME & MARKING GUIDELINES

TOPIC 7 – TRANSPORT IN PLANTS – Cambridge International AS Level Biology (9700)

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
1		Vascular bundles arranged in a peripheral ring surrounding central parenchyma pith (medulla); xylem located internally (facing pith), phloem located externally (facing cortex), separated by vascular cambium meristem; sclerenchyma fibres form a rigid bundle cap outside phloem [2].	[2]
1		Vascular tissue forms a central stele/cylinder; xylem forms a central star/cross (tetrarch) core with phloem clusters positioned between the arms; cylinder bounded by pericycle and outer endodermis; contrasts with stem where bundles are discrete, ringed, and surround wide central pith [2].	[2]
1		Stem: peripheral ring of lignified xylem and sclerenchyma resists lateral bending/shearing forces caused by wind (acts like cylindrical scaffolding); Root: central solid xylem core resists vertical longitudinal pulling/tensile strain as aerial parts sway or when pulled upward [2].	[2]
2		Xylem is located on the adaxial (upper) side; phloem is located on the abaxial (lower) side; reflects continuous divergence of stem vascular bundles into leaf traces where inner xylem extends into upper leaf surface and outer phloem extends into lower surface [2].	[2]
2		Palisade: columnar, vertically elongated cells tightly packed with minimal air spaces, containing ~80% of chloroplasts oriented for maximum light interception; Spongy: rounded, loosely packed cells with large moist intercellular air spaces facilitating rapid gas diffusion (CO ₂ , O ₂ , water vapour) [2].	[2]
2		Water exits xylem into mesophyll cell walls via osmosis/capillarity; evaporates from wet cell wall surfaces into intercellular air spaces; water vapour diffuses down a water potential gradient through open stomatal pores into the drier external atmosphere (transpiration) [2].	[2]
3		Autolysis of protoplast (cell contents including nucleus, cytoplasm, and vacuole break down, leaving a hollow empty lumen); breakdown/perforation of end walls between vertically stacked vessel elements, forming a continuous capillary tube with zero end resistance [2].	[2]
3		Lignin impregnates cellulose matrix, conferring immense mechanical rigidity and high tensile strength; prevents inward collapse (buckling/implosion) of vessel walls under extreme tension (negative hydrostatic pressure) generated by rapid transpiration pull [2].	[2]
3		Bordered pits are unlignified gaps in secondary walls where thin permeable primary wall (pit membrane) remains; allow lateral water movement between adjacent parallel vessels, providing bypass pathways around embolisms (air bubbles) caused by cavitation [2].	[2]
4		Peripheral cytoplasm with degeneration of nucleus, ribosomes, vacuole, and Golgi apparatus, leaving an open central lumen; end walls modified into perforated sieve plates with callose-lined pores allowing bulk flow of sap between elements [2].	[2]
4		Sieve tube elements lack nuclei/ribosomes and cannot synthesize essential proteins/enzymes, relying on companion cells for metabolic maintenance; companion cells contain prominent active nuclei, abundant mitochondria (ATP synthesis for proton pumping), and dense ribosomes [2].	[2]
4		Plasmodesmata (densely clustered symplastic cytoplasmic bridges); allow rapid passive symplastic diffusion of sucrose and ATP from companion cell cytoplasm into sieve tube lumen down concentration gradients [2].	[2]
5		Apoplast: water moves through porous cellulose cell walls and intercellular spaces via mass flow/capillarity with minimal resistance; Symplast: water moves through living cytoplasm and plasmodesmata via osmosis/diffusion across plasma membranes with higher resistance [2].	[2]
5		Composed of suberin (waxy, hydrophobic lipid-derived polymer) deposited in radial and transverse walls; completely blocks apoplastic route; forces water and dissolved ions across selectively permeable cell-surface membrane of endodermal cells into symplast [2].	[2]
5		Allows selective uptake and active regulation of essential mineral ions by carrier/channel proteins; prevents toxic/unwanted solutes and soil pathogens from entering xylem; maintains root pressure by preventing passive back-leakage of ions from stele into cortex [2].	[2]
6		Water evaporates from wet mesophyll cell walls, retreating into tiny pores between cellulose microfibrils; creates high surface tension and curved menisci; meniscus surface tension exerts negative hydrostatic pressure (suction/tension), pulling water from adjacent xylem vessels [2].	[2]
6		Cohesion: attraction between polar water molecules via hydrogen bonds, maintaining an unbroken, continuous water column with high tensile strength; Adhesion: attraction between water molecules and hydrophilic cellulose/lignin in xylem walls via hydrogen bonds, supporting column weight [2].	[2]
6		Midday: high transpiration creates extreme negative pressure/tension inside xylem vessels; vessel walls are pulled slightly inward by adhesion/tension, contracting trunk circumference; Night: stomata close, transpiration ceases, tension relaxes, vessel walls spring back, expanding trunk [2].	[2]
7		Cut shoot underwater at a slanting angle (prevents entry of air bubbles into xylem vessels which causes cavitation/air locks, and increases surface area for water absorption); assemble all connections underwater and apply petroleum jelly to joints to guarantee an airtight seal [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
7		Measure distance moved by air bubble meniscus along capillary tube (d) in millimetres using ruler; record time taken (t) in minutes using stopwatch; calculate volume = $\pi \times r^2 \times d$ (where r is internal radius of capillary); rate = volume / time [2].	[2]
7		Potometer measures water uptake, not water vapor loss directly; uptake exceeds transpiration because ~1-2% of water is chemically consumed as a substrate in photosynthesis (photolysis in thylakoids) and retained inside vacuoles to maintain cell turgor and cell expansion/growth [2].	[2]
8		Wind blows away the stagnant, moist boundary layer of humid air adhering to the leaf surface, steepening the water potential gradient between substomatal cavity and air; plateaus because stomatal pore resistance or mesophyll cell wall evaporation becomes the rate-limiting factor [2].	[2]
8		Higher external humidity increases the water potential of the ambient atmosphere (water potential becomes less negative); flattens/reduces the steepness of the water potential gradient between substomatal air space and outside air, slowing rate of diffusion [2].	[2]
8		Increasing temperature increases kinetic energy of water molecules, accelerating evaporation from mesophyll walls and diffusion through stomata; at extreme temperatures, excessive water loss causes loss of turgor in guard cells/stress response (abscisic acid ABA release), triggering stomatal closure [2].	[2]
9		Thick waxy cuticle on outer (abaxial) surface; sunken stomata restricted to inner grooves/pits; stiff epidermal trichomes (hairs); presence of specialised hinge (bulliform) cells (any 3) [2].	[2]
9		Trichomes and grooves trap a boundary layer of still, moist humid air inside the rolled leaf; prevents wind currents from blowing water vapour away; maintains high water potential outside stomata, reducing the water potential gradient and increasing diffusion distance [2].	[2]
9		During water drought/low humidity, large vacuolated hinge cells lose water by osmosis and lose turgor; shrinkage causes leaf to curl inward tightly, enclosing stomata in microclimate; when water is abundant, hinge cells take up water, become turgid, and flatten lamina for photosynthesis [2].	[2]
10		H ⁺ -ATPase proton pump actively hydrolyses ATP (ATP → ADP + Pi) to pump protons (H ⁺ ions) OUT of companion cell cytoplasm into cell wall/apoplast against their concentration gradient; creates high [H ⁺] and positive electrical potential in apoplast [2].	[2]
10		Protons diffuse back down their electrochemical gradient into companion cell via H ⁺ /sucrose cotransporter (symport protein); energy released by proton movement down gradient is coupled to cotransport of sucrose against its concentration gradient [2].	[2]
10		Cyanide/DNP inhibits ATP synthesis by oxidative phosphorylation in mitochondria; proton pumps cease operating; proton gradient dissipates; secondary active sucrose cotransport stops; phloem sap loading and mass flow cease completely [2].	[2]
11		Loading of sucrose into sieve tube lowers solute potential (Psi _s) and water potential (Psi); water enters sieve tube from adjacent xylem by osmosis down water potential gradient; rigid cellulose walls cannot expand, creating high hydrostatic pressure (turgor pressure) [2].	[2]
11		Sucrose is unloaded from sieve tubes into sink cells (for respiration or conversion into insoluble starch); raises water potential inside sieve tube; water exits sieve tube by osmosis into sink tissues/xylem, decreasing volume and lowering hydrostatic pressure [2].	[2]
11		Mass/bulk flow of water and dissolved solutes moves down the hydrostatic pressure gradient (delta-P) from source to sink; Evidence: (1) sap exudes under pressure from severed aphid stylets; (2) measured phloem translocation velocity (0.5-1.0 m h ⁻¹) far exceeds rate of diffusion [2].	[2]
12		Phloem sap is under high positive hydrostatic pressure generated by active loading and osmosis at source; laser severing leaves open micro-capillary without mechanical crushing that triggers normal wound response (callose/P-protein plugging) [2].	[2]
12		Peak moves from 10 cm (t = 1 h) to 40 cm (t = 3 h); distance = 30 cm in 2 hours = 15 cm h ⁻¹ (0.15 m h ⁻¹); impossible by diffusion alone because simple diffusion of sucrose over 30 cm would take decades; proves mass flow driven by pressure gradient [2].	[2]
12		Sucrose is a non-reducing disaccharide with glycosidic bond between anomeric carbons of glucose and fructose; chemically unreactive and cannot be metabolised directly during transit (unlike reducing glucose); carries twice the energy/carbon per molecule as hexose, maximising transport efficiency [2].	[2]
13		Cork/bark (periderm), cortex parenchyma, and phloem tissue (sieve tubes and companion cells) are excised; xylem (wood) remains completely intact in the interior [2].	[2]
13		Downward translocation of assimilates from photosynthesising source leaves is physically blocked at the girdle; sucrose, amino acids, and plant hormones (auxins) accumulate above the ring; increased osmotic solute concentration causes cell expansion, tissue swelling, and adventitious rooting [2].	[2]
13		Xylem vessels in the central wood remain undamaged, allowing transpiration pull to supply water and mineral ions continuously to leaves; tree eventually dies because roots below the girdle are deprived of sucrose, cannot carry out aerobic respiration, and cannot actively absorb mineral ions [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
14		Xylem: dead cells with no protoplast/cytoplasm, heavily lignified secondary walls, perforated/absent end walls; Phloem: living cells with peripheral cytoplasm (no nucleus), cellulose non-lignified walls, perforated sieve plates with callose pores [2].	[2]
14		Xylem: transpiration pull creates tension / negative hydrostatic pressure (-P), pulling water upward via cohesion-tension; Phloem: active solute loading creates positive hydrostatic turgor pressure (+P), driving sap by mass flow down a pressure gradient [2].	[2]
14		Xylem: entirely passive, powered by solar thermal energy driving evaporation of water from leaves; Phloem: active process requiring cellular metabolic ATP synthesised by companion cell mitochondria for proton pumping (chemiosmotic loading) [2].	[2]
15		Long, thin tubular cytoplasmic extension (root hair) drastically increases surface area-to-volume ratio; thin cellulose primary cell wall provides short diffusion distance; numerous mitochondria supply ATP for active transport [2].	[2]
15		Mineral ions are present in low concentrations in soil solution compared to root hair cytoplasm; must be transported against steep concentration gradients via specific carrier/cotransporter proteins using ATP; water enters passively by osmosis down water potential gradient [2].	[2]
15		Active accumulation of ions lowers solute potential (Ψ_s) and overall water potential (Ψ) inside root hair cytoplasm and vacuole; creates a steep water potential gradient between soil solution (less negative Ψ) and cell interior (more negative Ψ), driving rapid osmosis [2].	[2]
16		Inner cell wall facing the stoma is thick and inelastic, whereas outer wall is thin and flexible; cellulose microfibrils are arranged radially (radial micellation); when turgid, thin outer walls bulge outward, pulling thick inner walls apart to widen pore [2].	[2]
16		Light stimulates H^+ -ATPase pumps in guard cell membrane, pumping H^+ out; inside becomes negatively charged, opening voltage-gated K^+ channels; K^+ and Cl^- ions enter, lowering Ψ inside guard cells; water enters by osmosis, cells become turgid, pore opens [2].	[2]
16		Drought stimulates root ABA synthesis; ABA binds to guard cell receptors, opening Ca^{2+} channels; Ca^{2+} influx depolarises membrane, stimulating rapid efflux of K^+ and Cl^- ions; water potential rises, water exits by osmosis, guard cells lose turgor and pore closes [2].	[2]
17		Source: plant organ that produces more assimilate/sucrose than it consumes (net exporter); Sink: plant organ that consumes or stores assimilates (net importer); Summer potato: Source = mature photosynthesising green leaves; Sink = developing root tubers / growing shoot tips [2].	[2]
17		Autumn: tuber acts as sink, importing sucrose and converting it into insoluble starch for storage; Spring: starch is hydrolysed by amylase into maltose and glucose, converted to sucrose; sucrose is actively loaded into phloem to supply growing buds, making tuber a source [2].	[2]
17		Young leaves have high rates of cell division and expansion, lack fully developed chloroplasts, and respire more carbohydrate than they fix (net import of sucrose); as leaves expand and develop mature chloroplasts, photosynthesis exceeds respiration, becoming net exporters [2].	[2]
18		Endodermal cells actively secrete mineral ions into xylem vessels of root stele using ATP; lowers xylem water potential; water enters xylem from cortex by osmosis; Casparian strip prevents backflow; water accumulates in rigid xylem, creating positive root pressure [2].	[2]
18		Maximum root pressure measured is only ~100–200 kPa, capable of pushing water only 10–20 metres; tall trees require >2.0–3.0 MPa of pressure to overcome gravity and wall friction over 100 metres; some trees (e.g. conifers) show no root pressure [2].	[2]
18		Guttation: exudation of liquid water droplets from vein margins via hydathodes caused by root pressure under 100% humidity/night; Dew: condensation of water vapour from atmospheric air onto cold leaf surfaces (not internal xylem sap) [2].	[2]
19		High negative tension causes water to reach its metastable tensile limit or draws in micro-air bubbles; water column snaps (cavitation); air seeding: large pressure difference across pit membrane pulls air through pit pores into neighbouring functional vessel, spreading embolism [2].	[2]
19		Sigmoidal curve where PLC increases as water potential becomes increasingly negative; P50 is the xylem water potential (negative pressure) at which 50% of maximum hydraulic conductance is lost, serving as universal benchmark of drought vulnerability [2].	[2]
19		Narrower vessel diameters (higher capillary safety, lower risk of freeze-thaw cavitation); smaller pit membrane pores (restricts air seeding up to higher pressure differentials); redundant, highly branched vessel networks with abundant tracheids acting as safety buffers [2].	[2]
20		Diffusion through small pores is proportional to pore perimeter rather than area (edge effect); water molecules fan out in hemispherical vapor shells at pore edges, maintaining a steeper concentration gradient ($\Delta C / \Delta x$) around perimeters [2].	[2]
20		Stomatal conductance (g_s) is permeability determined by stomatal aperture, controlled actively by guard cell turgor/ABA; boundary layer conductance (g_b) is determined by thickness of still air layer, governed passively by wind speed and leaf geometry [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
20		WUE = mass of CO ₂ fixed / mass of water transpired; C4 plants concentrate CO ₂ internally using PEP carboxylase (no affinity for O ₂ , no photorespiration); can maintain smaller stomatal apertures to fix equivalent CO ₂ , drastically cutting transpiration loss [2].	[2]
21		Tracheids: elongated, spindle-shaped cells with tapered overlapping ends, possessing no perforations (water must pass through pit membranes); Vessels: wider, cylindrical cells stacked end-to-end with open/perforated end plates [2].	[2]
21		Vessels offer much higher hydraulic conductance (flow proportional to r ⁴ by Poiseuille's law), but an embolism incapacitates the entire vessel; Tracheids have lower flow rate but compartmentalise embolisms to single small cells, offering high safety [2].	[2]
22		Sudden pressure release causes surge of phloem sap; filamentous P-protein (phloem protein) rushes against sieve plate, physically plugging sieve pores; followed by rapid synthesis and enzymatic deposition of callose polymer around pore collars [2].	[2]
22		Callose is a beta(1→3)-glucan polysaccharide; in temperate perennials in autumn, callose is deposited over sieve plates, completely blocking pores during winter dormancy; in spring, callase enzyme hydrolyses callose, reopening sieve pores for translocation [2].	[2]
23		Xylem lumen → pit membrane → bundle sheath cell → mesophyll cell walls (apoplast) or cytoplasm (symplast) → evaporation from wet mesophyll cell walls → substomatal air space → diffusion through stomatal pore → outer air [2].	[2]
23		Extensive surface area of wet mesophyll cell walls is in direct contact with air spaces; continuous capillary replenishment of water from xylem maintains constant evaporation, equilibrating air spaces at near 100% relative humidity [2].	[2]
24		Stomata located exclusively on upper (adaxial) epidermis in contact with air (no stomata on submerged lower epidermis); large internal aerenchyma air chambers provide buoyancy (keeping leaf afloat) and allow O ₂ /CO ₂ diffusion [2].	[2]
24		Surrounded by water, so no requirement for long-distance water transport or drought resistance; water can be absorbed directly across entire plant surface; water provides mechanical buoyancy, eliminating requirement for extensive lignified support [2].	[2]
25		Apoplastic: sucrose exits mesophyll into cell wall and is actively pumped into companion cells via H ⁺ /sucrose cotransporters; Symplastic: sucrose diffuses directly through extensive branched plasmodesmata from mesophyll to intermediary cells without crossing cell walls [2].	[2]
25		Sucrose diffuses into intermediary companion cells through plasmodesmata; converted into larger raffinose/stachyose oligosaccharides; larger polymers cannot diffuse backward through narrow plasmodesmata into mesophyll, trapping sugar in phloem [2].	[2]
26		A thicker boundary layer increases the diffusion pathway length (delta-x) and holds high moisture; reduces the steepness of the water potential gradient between substomatal cavity and free atmosphere, dramatically reducing transpiration rate [2].	[2]
26		Large leaves have thicker boundary layers than small/divided leaves; surface trichomes (hairs) and rough cuticular ridges increase friction, trapping still air and preventing wind from eroding the boundary layer [2].	[2]
27		Protoxylem: narrow lumen, annular (ring) or spiral lignin thickenings; Metaxylem: much wider lumen, extensive reticulate or pitted lignification covering most of wall [2].	[2]
27		Protoxylem differentiates in regions undergoing rapid elongation; annular/spiral rings can stretch and uncoil like springs without snapping as surrounding tissues elongate; pitted walls would prevent elongation or rupture [2].	[2]
28		Stage 1: Casparian strip of suberin in radial/transverse walls; Stage 2: suberin lamella coats entire inner wall; Stage 3: thick cellulosic wall impregnated with lignin deposited on inner tangential and radial walls (U-shaped thickening) [2].	[2]
28		Passage cells are unthickened endodermal cells located opposite the protoxylem poles of the stele; remain thin-walled with only Casparian strip, providing low-resistance conduits for water and mineral ion transfer into xylem [2].	[2]
29		Xylem: inorganic nitrate (NO ₃ ⁻) and ammonium (NH ₄ ⁺), plus organic amides (glutamine, asparagine) synthesised in roots; Phloem: predominantly organic amino acids, amides, and small peptides (virtually zero free nitrate) [2].	[2]
29		Loaded via specific proton-amino acid cotransporters in companion cell plasma membrane, powered secondary-actively by the same H ⁺ electrochemical gradient generated by H ⁺ -ATPase pumps [2].	[2]
30		Solute potential (Psi _s): reduction in water potential due to dissolved solutes, always negative; Pressure potential (Psi _p): hydrostatic pressure exerted by cell wall on contents, positive in turgid cells (zero at incipient plasmolysis) [2].	[2]
30		Psi = -850 + 350 = -500 kPa; cell water potential (-500 kPa) is less negative (higher) than surrounding solution (-600 kPa); water moves by osmosis OUT of cell into sucrose solution down water potential gradient [2].	[2]
31		Water column under extreme tension snaps abruptly; rapid release of stored elastic energy in stretched water and deflected lignified vessel walls produces high-frequency vibrational sound waves (ultrasonic clicks) [2].	[2]
31		Peak afternoon solar radiation and temperature drive maximum transpiration rates while soil moisture is depleted; tensions reach their most negative values, exceeding cavitation thresholds of vulnerable vessels [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
32		Extensive invaginations/ridges of non-lignified primary cell wall projecting into cytoplasm, lined by plasma membrane; increases plasma membrane surface area by up to 10-fold [2].	[2]
32		Enables dense packing of thousands of additional H ⁺ -ATPase proton pumps and H ⁺ /sucrose cotransporter proteins per cell; dramatically accelerates rate of sucrose uptake from apoplast into phloem [2].	[2]
33		Chilling reduces kinetic energy, drastically increasing the viscosity of sucrose-rich phloem sap (resisting mass flow); inhibits respiration and ATP production in companion cells, impairing active loading [2].	[2]
33		Xylem conduits consist of dead empty tubes with no living membranes; transport is powered passively by solar evaporation from leaves (transpiration pull), not living stem metabolism [2].	[2]
34		Paint clear nail varnish on epidermis, peel off dried replica, mount on slide; count stomata in calibrated field of view using high power; calculate area of field ($\pi \times r^2$); stomatal density = count / area in mm ⁻² ; repeat and calculate mean [2].	[2]
34		Lower surface receives less direct solar radiation and is cooler, reducing evaporation; grooves and pits shelter stomata from wind, allowing humidity to build up and reducing the water potential gradient [2].	[2]
35		Transpiration is regulated biologically by stomatal aperture (guard cell turgor/ABA) and internal mesophyll resistance; evaporation from lake is purely physical, governed solely by vapor pressure deficit, wind, and net radiation [2].	[2]
35		Stomatal transpiration accounts for ~90% of total loss and is actively regulated; cuticular transpiration accounts for ~10% (direct diffusion through waxy cuticle) and is completely uncontrolled by the plant [2].	[2]
36		Air gases are compressible and have zero tensile strength; tension causes the air bubble to expand, filling the lumen and breaking water column continuity (vapor lock/air lock) [2].	[2]
36		Night root pressure generates positive hydrostatic pressure (+P), compressing air bubble and forcing gases to dissolve back into water; xylem parenchyma actively pump solutes into lumen, driving osmotic water influx to refill vessel [2].	[2]
37		Collenchyma: living cells with unevenly thickened primary cellulose and pectin walls (no lignin); Sclerenchyma: dead cells with heavily lignified, uniformly thickened secondary walls and empty lumens [2].	[2]
37		Collenchyma walls are non-lignified, plastic, and hydrated; provide flexible tensile support while retaining capacity to stretch and expand as stems elongate [2].	[2]
38		Sucrose diffuses down concentration gradient through plasmodesmata directly from sieve tubes into dividing meristematic cells, where it is rapidly hydrolysed and consumed in respiration and cell wall synthesis [2].	[2]
38		Sucrose exits phloem into cell wall apoplast; cell wall invertase hydrolyses sucrose into glucose and fructose; prevents sucrose re-entry and maintains steep concentration gradient favoring unloading [2].	[2]
39		Flow rate Q is proportional to radius to the fourth power (r^4); doubling sieve tube radius increases volumetric flow rate by $2^4 = 16$ -fold for an identical pressure gradient [2].	[2]
39		Sieve pores have much smaller radii than the main lumen; high drag/viscous friction occurs as sap funnels through constrictions; callose collars and P-protein filaments further narrow effective pore aperture [2].	[2]
40		Gravitational gradient = $100 \text{ m} \times 0.01 \text{ MPa m}^{-1} = 1.0 \text{ MPa}$; frictional resistance = 1.0 MPa ; total tension required = $-(1.0 + 1.0) = -2.0 \text{ MPa}$ (or -20 bar) [2].	[2]
40		At 100 m, extreme negative tension causes stomatal closure and limits xylem flow; coastal fog condensates on needles and is absorbed directly across cuticle into mesophyll, reversing water potential gradient and rehydrating crown [2].	[2]
41		The loss of water vapour from aerial parts of the plant (principally leaves); involving evaporation of water from wet mesophyll cell walls followed by diffusion of water vapour through stomata into the atmosphere down a water potential gradient [2].	[2]
42		The transport of organic assimilates (soluble organic solutes, predominantly sucrose and amino acids) in phloem sieve tube elements; moving from sources (regions of production) to sinks (regions of utilisation or storage); bidirectional [2].	[2]
43		Composed of suberin (waxy, hydrophobic lipid substance); completely blocks the apoplast pathway in radial and transverse endodermal walls, forcing water and dissolved mineral ions into the symplast [2].	[2]
44		Apoplast: interconnected system of non-living porous cell walls, intercellular spaces, and dead xylem lumens; Symplast: interconnected network of living cytoplasm linked cell-to-cell through plasmodesmata, bounded by plasma membranes [2].	[2]
45		Imparts high compressive and tensile mechanical rigidity, preventing collapse under extreme negative pressure (tension); renders secondary walls waterproof/impermeable, preventing leakage of sap [2].	[2]
46		Circular cavity in secondary lignified wall bridged by thin, unlignified primary wall and middle lamella (pit membrane); allows lateral water movement between adjacent vessels while preventing spreading of gas embolisms [2].	[2]
47		Synthesise ATP in abundant mitochondria to drive H ⁺ -ATPase proton pumps for active sucrose loading; synthesize proteins and enzymes that are transferred via plasmodesmata to maintain living sieve tube cytoplasm [2].	[2]
48		Marram grass (<i>Ammophila arenaria</i>) [or <i>Opuntia</i> / <i>Pinus</i>]; possessing rolled leaves with inward-facing stomata, thick waxy outer cuticle, sunken stomata in grooved pits, and epidermal trichomes (hairs) [2].	[2]

Q	PART	EXAMINER MARKING PRINCIPLES & ACCEPTED ANSWERS	MARKS
49		Perforated end wall containing enlarged plasmodesmata-derived sieve pores lined with callose; provides structural support preventing transverse wall collapse under pressure while offering minimal resistance to longitudinal mass flow [2].	[2]
50		Measures the rate of water uptake by a cut leafy shoot; based on the core assumption that rate of water uptake is approximately equal to the rate of transpiration under steady-state conditions [2].	[2]